# COL7560
# Assignment 1
# Data Collection using Active Measurements

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## 1 Introduction

This report analyzes the transport-layer characteristics of ChatGPT traffic, traditional web browsing, and video streaming. Using active measurement techniques and network shaping, we evaluate the impact of bandwidth and latency on application-level Quality of Experience (QoE).

## 2 Methodology: Application Data Collection

A total of 150 experiments were conducted, consisting of 50 independent iterations for each of the three application types. All experiments were performed under emulated network conditions with parameters controlled via a Linux `tc` (Traffic Control) shaper.

- **Bandwidth Range:** 100 – 4000 kbps
- **Emulated Latency:** 20 – 200 ms

### 2.1 Web Browsing Data Collection

The web browsing module measured the performance of standard HTTP/HTTPS traffic using a randomized set of URLs from a `sites.csv` repository.

**Automation Framework:** A Selenium-based Python script automated navigation and state management.

**Primary QoE Metric:** `Page Load Time (PLT)`, defined as the duration from the `driver.get()` command until the page reached a stable state.

**Telemetry:** Network-layer data was captured via `tcpdump` (PCAP), while application-level timings (DNS, TTFB) were logged using **BrowserMob Proxy** in HAR format.

**State Management:** Browsers utilized the `--incognito` flag and were restarted per iteration to ensure a clean cache state.

## 2.2 ChatGPT Data Collection

This module characterized the network footprint of generative AI streaming responses, specifically focusing on long-lived TLS/WebSocket sessions.

**Automation Framework:** A Puppeteer/Selenium suite simulated user interactions and technical prompt delivery.

**Primary QoE Metric:** `Response Time (RT)`, measured from the "Send" trigger until the disappearance of the UI "Stop" button (signaling the end of the text stream).

**Data Logging:** Focused on the analysis of `Server-Sent Events (SSE)` metadata and transport-layer characteristics.

## 2.3 Video Streaming Data Collection

The video module analyzed the footprint of modern `Adaptive Bitrate (ABR)` streaming using the DASH protocol.

**DASH Reference Player:** Controlled the `dash.js` player within a virtual display (`Xvfb`) for consistent rendering across 120-second intervals.

**Metrics:**

- `Average Resolution`: Polled every second via vertical pixel height.
- `Rebuffering Ratio`: The proportion of total session time where buffer occupancy fell below a 0.1s threshold.

**Analysis:** Real-time polling via the JavaScript API was used to aggregate mean resolution and total rebuffering frequency.

# 3 Data Analysis and Observations

We analyze how transport-layer features relate to application type and QoE using 150 experiments (50 per application). From each PCAP, the following flow-level features were extracted and aggregated per experiment:

- Mean inter-arrival time (iat_mean)
- Std. deviation of inter-arrival time (iat_std)
- Mean packet size (pkt_mean)
- Std. deviation of packet size (pkt_std)
- Throughput (throughput_bps)
- Flow duration

These were combined with application-specific QoE metrics:

- Web browsing: Page Load Time (PLT)
- ChatGPT: Total response time
- Video streaming: Rebuffering ratio

## 3.1 Application Classification and Feature Separability

To evaluate separability across applications, we applied dimensionality reduction using PCA and t-SNE.

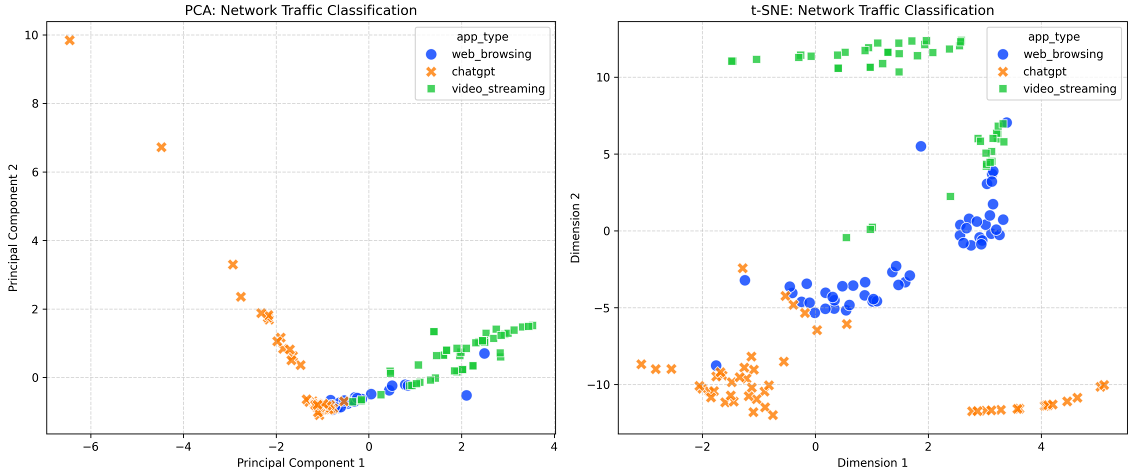


Figure 1: PCA and t-SNE visualization of network traffic features across different applications.

**Observations:**

- **ChatGPT Traffic:** Shows the highest degree of isolation due to streaming token delivery, which produces regular inter-arrival patterns.
- **Video Streaming:** Displays burst-driven behavior caused by segment downloads followed by buffering periods.
- **Web Browsing:** Shows partial overlap with both categories due to heterogeneous page content.

## 3.2 QoE Correlation Analysis

Below Figure shows Pearson correlation heatmaps between network features and QoE metrics.

**Web Browsing:** Page Load Time shows moderate positive correlation with throughput and packet statistics, and negative correlation with inter-arrival time features.

**ChatGPT:** Response time shows weak to moderate correlation with timing-related features, particularly `iat_std` and `iat_mean`.

**Video Streaming:** Rebuffering ratio shows relatively weak correlation with individual features, with slight sensitivity to delay variability.

**Overall Observation:** Timing-related features show more consistent influence on QoE than packet size alone, and throughput is not always a strong standalone predictor under controlled shaping conditions.

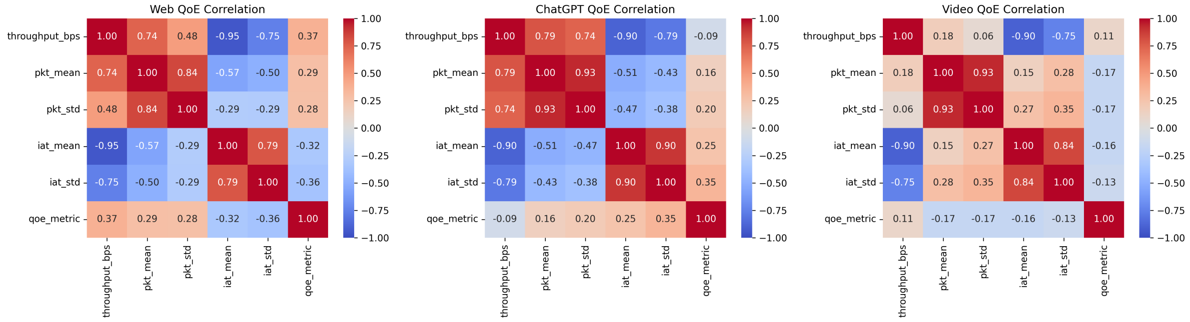


Figure 2: Correlation heatmaps between network traffic features and application-level QoE metrics.

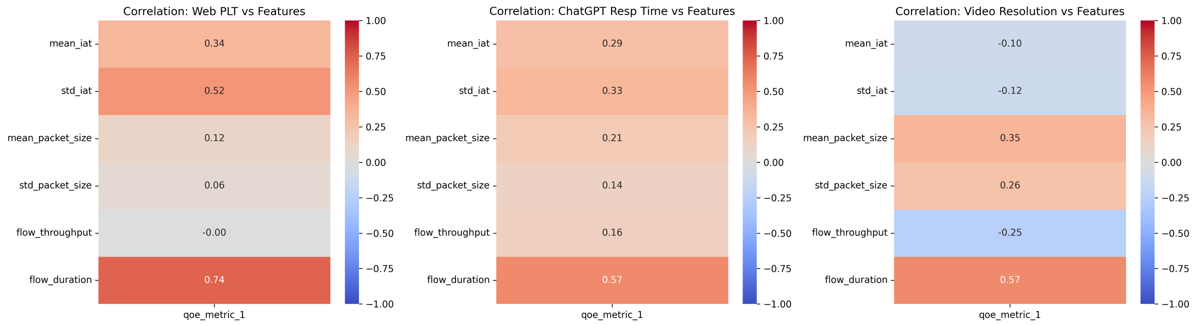


Figure 3: Correlation between network traffic features and application-level QoE metrics.